

Asymptotics for Guttman’s Multiple-Group CFA Estimator

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Purpose

Dhaene and Rosseel [4] report that Guttman’s multiple-group method [5] is a surprisingly strong non-iterative CFA estimator in their simulations. This note records the asymptotic objects needed to evaluate that claim analytically against normal-theory ML and the moment-quadratic estimators already implemented in `magmaan`.

The phrase “multiple-group method” below is Guttman’s historical name for forming factor score groups from indicators. It is not multi-group SEM. Actual multi-group SEM enters only by stacking independent covariance blocks.

1 Sample Moment Setup

Let $y_i \in \mathbb{R}^p$ have mean μ_0 , covariance Σ_0 , and finite fourth moments. Write

$$\hat{\sigma} = \text{vech}(\hat{\Sigma}), \quad \sigma_0 = \text{vech}(\Sigma_0),$$

with the usual covariance influence rows

$$z_i = \text{vech}\{(y_i - \mu_0)(y_i - \mu_0)^\top - \Sigma_0\}.$$

Then

$$\sqrt{n}(\hat{\sigma} - \sigma_0) = n^{-1/2} \sum_{i=1}^n z_i + o_p(1), \quad \Gamma = \text{Var}(z_i).$$

Under multivariate normality, Γ reduces to the normal-theory covariance of $\text{vech}(\hat{\Sigma})$. Under misspecification or non-normality, Γ is the actual fourth-moment covariance.

For independent sample groups $b = 1, \dots, B$, stack $\hat{\sigma} = (\hat{\sigma}_1^\top, \dots, \hat{\sigma}_B^\top)^\top$. With $n_b/N \rightarrow \pi_b$,

$$\sqrt{N}(\hat{\sigma} - \sigma_0) \Rightarrow N\{0, \text{blockdiag}(\Gamma_b/\pi_b)\}.$$

All formulas below apply to this stacked vector after replacing Γ by the block diagonal covariance.

2 The Guttman Map

Fix a simple CFA loading pattern. Let $X \in \{0, 1\}^{p \times k}$ be the indicator-by-factor membership matrix, with one nonzero entry per row in the clean simple-structure case. Let $S(\sigma)$ be the symmetric matrix whose lower triangle is σ .

Guttman's method first replaces the diagonal of S by communalities. Write

$$h(\sigma) \in \mathbb{R}^p, \quad C(\sigma) = S(\sigma) - \text{diag}\{\text{diag } S(\sigma)\} + \text{diag}\{h(\sigma)\}.$$

For the Spearman-Harman rule used by lavaan and magmaan starts, the unclamped correlation-scale communality for indicator i in its factor block is

$$h_i^{\text{cor}}(R) = |P_i|^{-1} \sum_{(j, \ell) \in P_i} \frac{R_{ij} R_{i\ell}}{R_{j\ell}}, \quad h_i(\Sigma) = \Sigma_{ii} h_i^{\text{cor}}\{R(\Sigma)\},$$

where $P_i = \{(j, \ell) : j < \ell, j \neq i, \ell \neq i\}$. The production start routine clamps this value and falls back in singular cases; the smooth theory below assumes an interior point where no clamp or fallback is active. A different communality rule only changes $Dh(\sigma_0)$.

Define

$$A = CX, \quad B = X^\top CX, \quad D = \text{diag}\{B\}, \quad Q = D^{-1/2}.$$

Then

$$L = AQ, \quad P = QBQ, \quad K^\star = LP^{-1}.$$

If m_f is the marker indicator for factor f , set

$$M = \text{diag}(K_{m_1 1}^\star, \dots, K_{m_k k}^\star).$$

The marker-scaled Guttman estimates are

$$\Lambda_G = K^\star M^{-1}, \quad \Phi_G = MPM, \quad \psi_G = \text{diag}\{S - \Lambda_G \Phi_G \Lambda_G^\top\}.$$

The reported parameter map is a fixed selection

$$\tau_G(\sigma) = [\text{sel}_\Lambda(\Lambda_G)^\top, \text{vech}(\Phi_G)^\top, \psi_G^\top]^\top.$$

This is the version closest to the current `guttman_start_values()` code: the full $K^\star$ is computed, but only the model's allowed loading cells are read out. Dhaene and Rosseel mention a constrained quadratic program for enforcing simple structure. If that projection is used instead, the same delta method applies after differentiating the fixed-active-set KKT system.

3 Guttman Influence Function

Let dS be a symmetric covariance perturbation and write all quantities at σ_0 . Let

$$dC = dS - \text{diag}\{\text{diag}(dS)\} + \text{diag}\{Dh[\sigma_0] d\sigma\}.$$

The differential chain is

$$\begin{aligned} dA &= dC X, \\ dB &= X^\top dC X, \\ dD &= \text{diag}\{dB\}, \\ dQ &= -\frac{1}{2} D^{-3/2} dD, \\ dL &= dA Q + A dQ, \\ dP &= dQ B Q + Q dB Q + Q B dQ, \\ dK^\star &= dL P^{-1} - L P^{-1} dP P^{-1}. \end{aligned}$$

Next

$$dM = \text{diag}(dK_{m_1 1}^\star, \dots, dK_{m_k k}^\star),$$

and

$$\begin{aligned} d\Lambda_G &= dK^\star M^{-1} - K^\star M^{-1} dM M^{-1}, \\ d\Phi_G &= dM P M + M dP M + M P dM, \\ d\psi_G &= \text{diag}\{dS - d\Lambda_G \Phi_G \Lambda_G^\top - \Lambda_G d\Phi_G \Lambda_G^\top - \Lambda_G \Phi_G d\Lambda_G^\top\}. \end{aligned}$$

Proposition 1 (Guttman delta method). *Assume Γ is finite, h is differentiable at σ_0 , the diagonal of B is positive, P is nonsingular, and all marker entries in M are nonzero. Then*

$$\sqrt{n}\{\tau_G(\hat{\sigma}) - \tau_G(\sigma_0)\} \Rightarrow N(0, \Omega_G), \quad \Omega_G = J_G \Gamma J_G^\top,$$

where $J_G = D\tau_G[\sigma_0]$ is the linear map defined by the differential chain above. The observation-level influence function is

$$\text{IF}_G(y_i) = J_G \text{vech}\{(y_i - \mu_0)(y_i - \mu_0)^\top - \Sigma_0\}.$$

Remark. The current magmaan implementation already has most of the numerical map in `src/estimate/start_guttman.cpp`. Turning it into an estimator is not hard. The open design choice is whether the estimator should be the clean smooth map above, the lavaan-style clamped start map, or the constrained projection variant used by Dhaene and Rosseel. For asymptotic comparisons, the clean map is the right first target.

4 Guttman Omega and Alpha

Let $w \in \mathbb{R}^p$ be fixed score weights. For a diagonal-residual CFA model, coefficient omega for the composite $w^\top y$ is

$$\omega(\Lambda, \Phi, \psi; w) = \frac{w^\top \Lambda \Phi \Lambda^\top w}{w^\top S w},$$

where S is the covariance matrix whose reliability is being summarized [9]. Applying this formula to the Guttman map gives

$$\omega_G(\sigma; w) = \frac{w^\top \Lambda_G \Phi_G \Lambda_G^\top w}{w^\top S(\sigma) w}.$$

This looks parameter-based, but the parameters cancel. Since

$$\Lambda_G \Phi_G \Lambda_G^\top = K^\star P (K^\star)^\top = L P^{-1} L^\top = C X (X^\top C X)^{-1} X^\top C,$$

we have the covariance-only form

$$\omega_G(\sigma; w) = \frac{w^\top C X (X^\top C X)^{-1} X^\top C w}{w^\top S(\sigma) w}. \quad (1)$$

Thus, once the grouping matrix X and communality rule $h(\sigma)$ are fixed, Guttman omega is a population covariance functional. It does not require iterative loadings, fitted residual variances, or marker choices. It is not assumption-free: X , h , and the nonsingularity of $X^\top C X$ are part of the definition.

For one factor, $X = \mathbf{1}$. Put

$$a = C\mathbf{1}, \quad b = \mathbf{1}^\top C\mathbf{1}, \quad T = \mathbf{1}^\top S\mathbf{1}.$$

Then

$$\omega_G(w) = \frac{(w^\top a)^2}{b w^\top S w}.$$

For the unit-weight total score $w = \mathbf{1}$, this collapses further:

$$\omega_G = \frac{\mathbf{1}^\top C \mathbf{1}}{\mathbf{1}^\top S \mathbf{1}} = \frac{\sum_{i \neq j} S_{ij} + \sum_i h_i}{T} = 1 - \frac{\sum_i (S_{ii} - h_i)}{T}. \quad (2)$$

Here h_i is the communality placed on item i 's diagonal in C . For the Spearman-Harman/Guttman rule above, in the one-factor case,

$$\begin{aligned} h_i &= \binom{p-1}{2}^{-1} \left\{ \sum_{1 \leq j < \ell \leq p} \frac{S_{ij} S_{i\ell}}{S_{j\ell}} - (p-1) S_{ii} \right\} \\ &= \{2 \binom{p-1}{2}\}^{-1} \left\{ \sum_{j, \ell=1}^p \frac{S_{ij} S_{i\ell}}{S_{j\ell}} - \sum_{j=1}^p \frac{S_{ij}^2}{S_{jj}} - 2(p-1) S_{ii} \right\}. \end{aligned}$$

This is the covariance-scale form of the correlation expression $S_{ii} R_{ij} R_{i\ell} / R_{j\ell}$: the item variance factors cancel, and the unrestricted pair sum contains $p-1$ pairs involving item i , each reducing to S_{ii} . The full double sum counts each unordered off-diagonal pair twice. Its subtraction removes the diagonal contribution $\sum_j S_{ij}^2 / S_{jj}$ and the $2(p-1)$ off-diagonal terms involving item i , each reducing to S_{ii} . In a multi-factor simple-structure block, replace p by the block size p_f and sum only within the block. Under an exact congeneric one-factor population, this population rule recovers $h_i = \lambda_i^2 \phi$. So the one-factor Guttman-derived omega is especially simple: replace each item variance by its Guttman communality, sum all entries, and divide by the observed total-score variance.

Cronbach's alpha for the same unit-weight score is [3]

$$\alpha = \frac{p}{p-1} \left(1 - \frac{\text{tr}(S)}{\mathbf{1}^\top S \mathbf{1}} \right) = \frac{p}{p-1} \frac{\sum_{i \neq j} S_{ij}}{T}. \quad (3)$$

Equations (2) and (3) show the contrast. Alpha extrapolates reliability from the off-diagonal covariance average. Guttman omega uses the same off-diagonal covariance sum, but adds an explicit diagonal common-variance estimate $\sum_i h_i$.

Under an exact one-factor congeneric population,

$$\Sigma = \lambda \lambda^\top \phi + \text{diag}(\psi),$$

and under a communality rule that recovers $h_i = \lambda_i^2 \phi$,

$$\omega_G = \frac{\phi(\mathbf{1}^\top \lambda)^2}{\phi(\mathbf{1}^\top \lambda)^2 + \mathbf{1}^\top \psi},$$

which is the usual omega total for the unit-weight sum score. In the same case,

$$\omega_G - \alpha = \frac{\phi\{p \sum_i \lambda_i^2 - (\sum_i \lambda_i)^2\}}{(p-1)T} \geq 0,$$

with equality exactly when the keyed loadings are equal. Thus alpha and Guttman omega agree in the tau-equivalent case; with congeneric loadings, alpha is lower because it has no direct diagonal common-variance estimate.

Away from the exact one-factor model, both quantities remain smooth covariance functionals, but they target different summaries. Alpha needs no loading pattern and is easiest to compute. Guttman omega is still non-iterative, but it imports the CFA grouping pattern and the chosen communality estimator. Its large-sample variance follows immediately by the delta method:

$$\sqrt{n}\{\hat{\omega}_G - \omega_G\} \Rightarrow N\{0, \nabla \omega_G(\sigma_0)^\top \Gamma \nabla \omega_G(\sigma_0)\},$$

and alpha has the same form with $\nabla \alpha(\sigma_0)$.

Relation to Hancock and An (2020). Hancock and An [6] independently proposed a closed-form single-factor omega and evaluated it in a large simulation against the CFA-based estimate. Their loading uses Spearman's [8] ratio-of-sums rule,

$$\hat{\lambda}_i = \left\{ \frac{\sum_{j < k} S_{ij} S_{ik}}{\sum_{j < k} S_{jk}} \right\}^{1/2}, \quad \hat{\omega} = \frac{(\sum_i \hat{\lambda}_i)^2}{T}, \quad j, k \neq i.$$

Under an exact one-factor population this coincides with (2): both recover $\hat{\lambda}_i^2 = \lambda_i^2 \phi$ and hence the same omega. Off the one-factor model the two differ. The communality h_i above is an *average* of the per-triple ratios $S_{ij} S_{ik} / S_{jk}$, whereas Hancock and An pool numerators and denominators before dividing; pooling is the more numerically stable choice when some $S_{j\ell}$ are near zero, so for the single-factor unit-weight case their ratio-of-sums aggregation is preferred. The single-factor closed-form omega is therefore not new with this note (numerically confirmed: the two agree to machine precision on a one-factor population and diverge, e.g. 0.712 vs 0.749, on a two-factor covariance analyzed as one factor).

What (1) adds beyond Hancock and An, who restrict to a single factor, is the grouping matrix X and the weights w : a closed-form omega *total* for multi-factor, multi-group, and weighted-composite reliability. The natural next target is omega *hierarchical*: a second-stage centroid on the Guttman factor correlation Φ_G (a Schmid-Leiman general factor) yields closed-form general-factor loadings whenever $k \geq 3$, and hence a closed-form ω_h ; a non-proportional bifactor solution lies outside the simple-structure X and is not available by this route.

5 Guttman Omega Hierarchical

Omega hierarchical ω_h is the share of total score variance carried by a single general factor under a Schmid-Leiman [10] orthogonalization of a correlated k -factor model. Write $\Sigma = \Lambda\Phi\Lambda^\top + \Psi$ with the second-order factorization of the factor correlation $\Phi = \gamma\gamma^\top + \Psi_\varphi$, where $\gamma \in \mathbb{R}^k$ holds the loadings of the first-order factors on the general factor. With simple structure the general-factor item loadings are $\lambda_g = \Lambda\gamma$, and

$$\omega_h = \frac{(\mathbf{1}^\top \lambda_g)^2}{\mathbf{1}^\top S \mathbf{1}} = \frac{(\mathbf{1}^\top \Lambda \gamma)^2}{T}, \quad T = \mathbf{1}^\top S \mathbf{1}.$$

The first stage is the Guttman map already given: in the unit-variance metric the loadings are $K^* = CXB^{-1}D^{1/2}$ and the factor correlation is $P = D^{-1/2}BD^{-1/2}$, with $B = X^\top CX$, $D = \text{Diag } B$. The second stage applies the *same* centroid extraction to P : replace its unit diagonal by second-order communalities to form C_P , and read off

$$\gamma_P = \frac{C_P \mathbf{1}}{\sqrt{\mathbf{1}^\top C_P \mathbf{1}}}, \quad \lambda_g = K^* \gamma_P.$$

Then $\lambda_g \lambda_g^\top = K^* \gamma_P \gamma_P^\top K^{*\top}$ is the general-factor item covariance and

$$\omega_h(\sigma) = \frac{(\mathbf{1}^\top K^* \gamma_P)^2}{T} = \frac{(a^\top B^{-1} D^{1/2} C_P \mathbf{1})^2}{(\mathbf{1}^\top C_P \mathbf{1}) T}, \quad a = X^\top C \mathbf{1}. \quad (4)$$

This needs $k \geq 3$ first-order factors, since the second-stage triples are over factors. It is the Schmid-Leiman *proportional* general factor; a free (non-proportional) bifactor is outside the simple-structure X .

In Hancock and An's [6] elementary form this is their ratio-of-sums rule applied at two levels. Within block f , $\hat{\lambda}_i = \{\sum_{j < \ell \in f} S_{ij} S_{i\ell} / \sum_{j < \ell \in f} S_{j\ell}\}^{1/2}$ and $s_f = \sum_{i \in f} \hat{\lambda}_i$; the factor correlations are $\hat{\varphi}_{fg} = \sum_{i \in f, j \in g} S_{ij} / (s_f s_g)$;

the same rule one level up gives $\hat{\gamma}_f = \{\sum_{g < h \neq f} \hat{\varphi}_{fg} \hat{\varphi}_{fh} / \hat{\varphi}_{gh}\}^{1/2}$; and $\hat{\omega}_h = (\sum_f \hat{\gamma}_f s_f)^2 / T$. On the exact higher-order manifold this two-level form and (4) coincide and recover the true ω_h .

Influence function. $\omega_h = g(\sigma)$ is a smooth covariance functional, so by the delta method $\sqrt{n}\{\hat{\omega}_h - \omega_h\} \Rightarrow N(0, \nabla g^\top \Gamma \nabla g)$ with $\text{IF}(y_i) = \nabla g(\sigma_0)^\top z_i$. The gradient composes the matrix differentials

$$\begin{aligned} dC &= dS - \text{Diag}(\text{diag } dS) + \text{Diag}(dh), \\ dB &= X^\top dC X, \\ dP &= D^{-1/2} dB D^{-1/2} - \frac{1}{2}(\Delta P + P \Delta), \quad \Delta = D^{-1} \text{Diag}(dB), \\ dC_P &= dP - \text{Diag}(\text{diag } dP) + \text{Diag}(dh^\varphi), \end{aligned}$$

where dh and dh^φ are the first- and second-stage communality differentials, the only stage-specific pieces; a different communality rule changes nothing else. Vectorizing, each stage is Kronecker-linear in $\text{vech } dS$: the lift is $\text{vec } dB = (X^\top \otimes X^\top) \text{vec } dC$, the inverse uses $d(B^{-1}) = -B^{-1}(dB)B^{-1}$, and every scalar contraction uses $a^\top(dM)b = (b^\top \otimes a^\top) \text{vec } dM$. With $N = a^\top B^{-1} D^{1/2} C_P \mathbf{1}$,

$$\begin{aligned} dN &= (X^\top dC \mathbf{1})^\top B^{-1} D^{1/2} C_P \mathbf{1} - a^\top B^{-1} (dB) B^{-1} D^{1/2} C_P \mathbf{1} \\ &\quad + a^\top B^{-1} (dD^{1/2}) C_P \mathbf{1} + a^\top B^{-1} D^{1/2} (dC_P) \mathbf{1}, \end{aligned}$$

and the quotient rule on $\omega_h = N^2 / [(\mathbf{1}^\top C_P \mathbf{1}) T]$ gives $\nabla \omega_h$. Only the lift $X^\top \otimes X^\top$ and the two communality blocks distinguish this from the single-factor case, so Γ 's vech structure passes straight into $\Omega = \nabla \omega_h^\top \Gamma \nabla \omega_h$. Numerically, (4) recovers the true ω_h to machine precision on a higher-order population ($k = 3$ blocks of three items, $\omega_h = 0.6134277$), and its delta-method standard error matches a Monte Carlo standard deviation to within one percent.

6 Moment-Quadratic Estimators

Let $m(\theta)$ be the model-implied covariance moment vector and

$$r(\theta, \sigma) = m(\theta) - \sigma.$$

A fixed-weight ULS/GLS/WLS estimator minimizes

$$q_W(\theta, \sigma) = \frac{1}{2} r(\theta, \sigma)^\top W r(\theta, \sigma),$$

with pseudo-true value θ_W . Let

$$\Delta = \frac{\partial m(\theta_W)}{\partial \theta^\top}, \quad H_W = \Delta^\top W \Delta + \sum_{\ell} \{W r(\theta_W, \sigma_0)\}_{\ell} \frac{\partial^2 m_{\ell}(\theta_W)}{\partial \theta \partial \theta^\top}.$$

The second term is the misspecification curvature term. It vanishes at exact fit and is easy to miss if one silently substitutes Fisher or Gauss-Newton information.

For fixed W ,

$$\text{IF}_W(y_i) = H_W^{-1} \Delta^\top W z_i, \quad \Omega_W = H_W^{-1} \Delta^\top W \Gamma W \Delta H_W^{-1}.$$

If $W = W(\hat{\sigma})$, the weight influence is first-order under misspecification:

$$\text{IF}_{W(\hat{\sigma})}(y_i) = H_W^{-1} \Delta^\top W z_i - H_W^{-1} \Delta^\top \{DW[\sigma_0]z_i\} r(\theta_W, \sigma_0).$$

This second term drops out at exact fit. It is therefore most important in the setting the user asked about: fixed model misspecification with GLS/WLS style weights. This is also the term already handled by magmaan's continuous LS influence machinery when the caller asks for data-dependent weight influence.

The efficient covariance-moment estimator for a fixed covariance target uses $W = \Gamma^{-1}$, giving the familiar

$$\Omega_{\text{eff}} = (\Delta^\top \Gamma^{-1} \Delta)^{-1}$$

only at exact fit or when the curvature and weight-influence terms are absent. Under misspecification the observed bread, not the Fisher simplification, is the general object.

7 Normal-Theory ML

For covariance-only NT estimation,

$$F_{\text{NT}}(\theta, \sigma) = \log |\Sigma_\theta| + \text{tr}\{S(\sigma)\Sigma_\theta^{-1}\} - \log |S(\sigma)| - p.$$

The term $-\log |S|$ is irrelevant for the first-order condition. At the pseudo-true point θ_{NT} , define

$$U_a(\theta, \sigma) = \text{tr} \left[\left\{ \Sigma_\theta^{-1} - \Sigma_\theta^{-1} S(\sigma) \Sigma_\theta^{-1} \right\} \frac{\partial \Sigma_\theta}{\partial \theta_a} \right].$$

Let $H_{\text{NT}} = \partial U / \partial \theta^\top$ be the observed pseudo-true bread. For a covariance perturbation dS ,

$$\{G_{\text{NT}} d\sigma\}_a = -\text{tr} \left[\Sigma_\theta^{-1} dS \Sigma_\theta^{-1} \frac{\partial \Sigma_\theta}{\partial \theta_a} \right].$$

Thus

$$\text{IF}_{\text{NT}}(y_i) = -H_{\text{NT}}^{-1} G_{\text{NT}} z_i.$$

When the normal model is correct this is the usual efficient likelihood influence function. Under non-normality or fixed structural misspecification, the sandwich $H^{-1} G \Gamma G^\top H^{-1}$ is the robust covariance around the NT pseudo-true parameter.

8 What Efficiency Means Under Misspecification

Under fixed misspecification, Guttman, NT, ULS, GLS, and WLS generally estimate different functionals of the same population covariance. A variance ratio is therefore not automatically an MSE ratio.

For a scalar contrast $a_e\{\tau_e(P)\}$, the variance-only ratio is

$$\rho_{G/e} = \frac{\nabla a_G^\top \Omega_G \nabla a_G}{\nabla a_e^\top \Omega_e \nabla a_e}.$$

This is meaningful when the contrast names the same scientific quantity and the pseudo-targets are close enough that first-order variance is the object of interest.

For fixed misspecification, a finite- n target-risk comparison is more honest:

$$R_e(n; A, \tau_0) = \{\tau_e(P) - \tau_0\}^\top A \{\tau_e(P) - \tau_0\} + \frac{1}{n} \text{tr}(A \Omega_e).$$

Dhaene and Rosseel's MSE comparisons are of this type, with A selecting groups of parameters and τ_0 the data-generating parameter vector. An analytic experiment should report both R_e and the variance-only ratios.

A third option is local misspecification:

$$\sigma_n = \sigma\{\theta_0\} + n^{-1/2} \delta.$$

Then bias and variance both contribute at first order, and the influence maps above give closed-form local asymptotic risk. This is probably the cleanest paper-level efficiency comparison.

9 Proposed magmaan Experiment

The experiment should be deterministic and population based first, with Monte Carlo used only to validate the formulas.

1. Estimator maps. Implement a frontier Guttman CFA estimator that returns $(\Lambda_G, \Phi_G, \psi_G)$ per covariance block using the clean map in this note. Reuse `cfa_block_layouts()` and `guttman_start_values()` as implementation scaffolding, but keep finite-sample clamps out of the analytic target.

2. Jacobians. Start with central finite differences of $\tau_G(\sigma)$ to verify the chain above. Then add the analytic differential for speed and numerical diagnostics. The Spearman-Harman communality derivative is a quotient-rule calculation through the covariance-to-correlation map.

3. Comparator estimators. For each population covariance, fit the analysis model at the population moments with NT, ULS, GLS, DWLS/WLS, and possibly empirical-ADF WLS. Use the current observed-bread sandwich machinery for NT and continuous LS. For data-dependent GLS/WLS weights, run both fixed-weight and weight-influence versions to show how much the misspecification term matters.

4. Designs. Replicate the Dhaene-Rosseel three-factor, three-indicator design with reliability 0.5/0.8, factor correlations 0.1/0.4, omitted cross-loadings, and omitted residual covariances. Add one larger block design with six to twelve indicators per factor, because Guttman’s computational appeal grows with p .

5. Distributional covariance. Evaluate Γ under three choices: normal theory, an analytic elliptical or independent-component heavy-tailed fourth-moment model, and empirical fourth moments from a very large simulated DGP draw. This separates structural misspecification from non-normal covariance noise.

6. Outputs. For each parameter group and selected scalar contrasts, report:

$$\text{diag}(\Omega_G)/\text{diag}(\Omega_e), \quad \text{tr}(A\Omega_G)/\text{tr}(A\Omega_e), \quad R_G(n; A, \tau_0)/R_e(n; A, \tau_0).$$

Also report the pseudo-target displacement $\tau_e(P) - \tau_0$, because under fixed misspecification this often dominates large- n MSE.

7. Validation. For a small subset, simulate many datasets, compute all estimators, and check that $n \widehat{\text{Cov}}(\hat{\tau}_e)$ matches Ω_e . For Guttman this is the main new validation. For NT and LS it should agree with existing misspecification-robust sandwich code. This is implemented as the `-validate` path in experiment 40: at the well-specified cell with 2000 replications and $N = 1000$, the Guttman all-parameter asymptotic SD matches simulation to within 1% (−0.8%), bracketed by the known-correct NT (−0.7%) and oracle-WLS (−1.0%) sandwiches, so the finite-difference Jacobian of the Guttman map is as accurate as those comparators.

10 Takeaways

Guttman’s estimator has straightforward regular asymptotics because it is a smooth covariance-summary map away from clamps, singular factor-score correlations, and zero marker loadings. Its influence function is simply the Jacobian of that map times the covariance influence row.

The comparison to NT, ULS, GLS, and WLS should not be framed as one universal efficiency ranking. Under correct normal specification NT/GLS have the usual optimality story. Under fixed misspecification, the relevant objects are pseudo-targets, observed breads, empirical meats, and weight influence. That is exactly the regime where an analytic magmaan experiment can be more informative than another Monte Carlo table.

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